

T2 Evolution Concept Disambiguation Synthesis

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

Date: May 15, 2026

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Series: T (Trilogy Ambiguity Reduction), Note T2.05 (#698)

Attribution

This work derives from and formalizes commitments across the CKS theory trilogy: *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026), and *Inter-Self Coordination via Shared Substrate / Full Aspect Integration* (Li, April 2026).

Abstract

Notes T2.01 through T2.04 (Series T, #694–#697) disambiguated four evolution concepts that appear across the CKS trilogy with scope-specific meanings: directed selection, action-feedback, mutation, and proposing substrate. This synthesis note states what those four disambiguations jointly established. Across three of the four concepts — directed selection, action-feedback, and proposing substrate — the same mechanism operates at Paper 2 and Paper 3 scope; Paper 3 extends through new inter-Self sources or optional governance configuration without replacing the mechanism. Mutation is the exception: the one evolution concept with no inter-Self scope expression, architecturally home-bound at every trilogy scope. The asymmetry between these two groups is not incidental — it is the direct consequence of the instinct/reasoning separation and the governance perimeter structure established in earlier Series T notes. This note states the pattern, the exception, and the asymmetry, and opens the transition to Series T Phase T3.

1. Purpose of this synthesis note

Series T Phase T2 addressed a specific adversary risk: that an adversary could claim Paper 3's evolution architecture is novel by treating the evolution concepts that appear in Papers 2 and 3 as separate inventions rather than as scope-specific expressions of the same mechanisms. Notes T2.01–T2.04 individually closed that risk for four concepts. This synthesis note — T2.05 — states what those four closures jointly establish: a consistent pattern across three reasoning-layer mechanisms, an architectural exception for the one instinct-layer mechanism, and the asymmetry between them as a structural consequence that no individual T2 note stated in full.

The synthesis serves two purposes. First, it provides a consolidated prior-art reference for the T2 pattern, allowing downstream notes to reference it without re-deriving it. Second, it names the asymmetry as architecturally fundamental — the T2 subseries' primary contribution beyond the individual disambiguations.

2. Four T2 disambiguations in summary

T2.01 — Directed selection. The modify right (Paper 1, Claim 3) is the authority basis for governance-authorized DNA changes at every trilogy scope. Paper 3's DNA absorption — governance-authorized selective intake of DNA content from a FAI event into a home Self's DNA layer — is directed selection with an inter-Self source. Absorption does not introduce a new mechanism; it extends directed selection to receive inputs from across the inter-Self perimeter.

T2.02 — Action-feedback. Three distinct concepts share this term: action-layer storage as a substrate layer (Papers 1 and 2), the action-feedback pipeline as the loop from recorded action evidence to governed DNA refinement (Paper 2), and the action-feedback locus as the inter-Self delivery pathway through which FAI outcomes enter a home Self's action layer (Paper 3). The locus is the entry point to the pipeline, not the pipeline itself. The mechanism is the same at Papers 2 and 3 scope; Paper 3 adds an inter-Self input channel.

T2.03 — Mutation. Instinct evolution — LLM version changes and substrate-platform upgrades that sharpen the fast-pattern layer without changing DNA content — has no inter-Self scope expression anywhere in the trilogy. Paper 3's instinct-layer exclusion at the inter-Self perimeter is an explicit architectural commitment: instinct-layer content does not cross the inter-Self perimeter. Mutation is home-bound at every scope by design, not by omission.

T2.04 — Proposing substrate. The LLM-mediated mechanism that generates proposals for DNA changes from action evidence is the same mechanism at Papers 2 and 3 scope. Paper 3 adds optional governance configuration (D2.59 Configurations A/B/C) that enables the proposing substrate to distinguish FAI-origin evidence from home-origin evidence when generating proposals. The default mode is unchanged. Configuration is a governance quality extension, not a new proposing mechanism.

3. The T2 pattern: reasoning-layer mechanisms extend to inter-Self scope

Across three of the four T2 concepts, the same structural pattern holds. The mechanism established at Paper 2 scope extends to Paper 3 scope by receiving new inputs from across the inter-Self perimeter, or by accepting optional governance configuration that adapts its behavior to inter-Self sources. The mechanism does not change; what changes is the scope from which it receives inputs and the governance surface available for configuring how it handles those inputs.

This is the consistent T2 pattern: **same mechanism, new inter-Self sources or configuration.** It applies to directed selection (new source: DNA content from FAI events), action-feedback (new source: action-layer content from FAI outcomes), and proposing substrate (new configuration: optional FAI-origin awareness). All three operate on reasoning-layer content — DNA-layer and action-layer substrate — under human governance and outside the LLM.

4. The mutation exception: architecturally home-bound

Mutation does not follow the T2 pattern. There is no inter-Self extension for instinct evolution, no optional configuration that makes LLM-layer content available across the inter-Self perimeter, and no FAI locus that delivers instinct-layer content from one Self to another. This is not a gap in the pattern; it

is the architectural commitment Paper 3 makes explicit and Papers 1 and 2 imply.

The exception is as architecturally significant as the pattern. An adversary observing that directed selection, action-feedback, and proposing substrate all extend to inter-Self scope might argue that mutation should extend similarly, and that Paper 3's restriction is a novel negative constraint. T2.03 closes that argument: the restriction is not novel because it follows directly from the governance perimeter and substrate structure established in Papers 1 and 2. The exception is inherited, not invented.

Mutation's home-bound character has a safety consequence: undirected instinct-layer change cannot propagate across organizational boundaries through any inter-Self exchange mechanism. Cells evolve their instinct layers within their own governance perimeter; no mechanism in the trilogy propagates that undirected change outward.

5. The asymmetry is architecturally fundamental

The gap between the three reasoning-layer mechanisms and the instinct-layer mechanism is the T2 subseries' structural finding. It is not a consequence of terminology choices or paper organization. It is the consequence of two architectural commitments that earlier Series T notes established.

T1.03 (substrate/aspect disambiguation) established the distinction between the governance substrate — the coordination medium that carries DNA-layer and action-layer content and is designed to be authored, inspected, modified, and transported as governed substrate content — and the instinct layer that the LLM occupies, which is governed only indirectly through verification and routing rules that the governance substrate specifies.

T1.06 (governance perimeter) established that the instinct/reasoning separation is a within-Self governance boundary. Reasoning-layer content can, in principle, cross the inter-Self perimeter because it is governed substrate content. Instinct-layer content cannot, because it lives inside the LLM and is not governed as authored substrate content.

Together, these commitments produce the mutation exception: because mutation operates on instinct-layer content, and because instinct-layer content does not cross the inter-Self perimeter by architectural commitment, mutation has no inter-Self scope expression. The three reasoning-layer mechanisms extend because their content is governed substrate content. Mutation does not extend because its content is not. Removing the asymmetry would require revising commitments that Papers 1 and 2 defend as foundational, not merely adding a new mechanism.

6. Transition to T3

The T2 subseries addressed evolution concept disambiguation — concepts that name how a Self and its components change over time. The disambiguation work established that when the same evolution mechanism term appears in Papers 2 and 3, it names the same mechanism at different scopes, with Paper 3 adding inter-Self input channels or optional configuration but not new mechanisms.

Series T Phase T3 turns to architectural concept disambiguation. The concepts in scope — cell, aspect, Self, layer, and scope itself — are structural terms that appear across all three trilogy papers. Like the

evolution mechanism terms, they carry consistent meanings throughout the trilogy. Their scope-specific applications produce progressively larger coordination structures from the Paper 1 cell, with the same underlying architectural object instantiated at progressively larger perimeters.

T3's disambiguation question is: when the same structural term appears in different papers, what does it name at each scope, and what is the commitment that it is the same kind of thing? T2's answer for evolution mechanisms provides the template. T3 will show the same pattern holds for structural terms: one architectural primitive, scope-specific application.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

Related Series T notes

Li, W. (2026). *T2.01 — Directed Selection Disambiguation (#694)*. Series T, CKS trilogy derivation series.

Li, W. (2026). *T2.02 — Action-Feedback Disambiguation (#695)*. Series T, CKS trilogy derivation series.

Li, W. (2026). *T2.03 — Mutation Disambiguation (#696)*. Series T, CKS trilogy derivation series.

Li, W. (2026). *T2.04 — Proposing Substrate Disambiguation (#697)*. Series T, CKS trilogy derivation series.

How to cite this note

Li, W. (2026). *T2 Evolution Concept Disambiguation Synthesis*. May 15, 2026. Series T, Note T2.05 (#698). ORCID: 0009-0004-8065-3235.